

Plants, Part 1: Structure & Transpiration

Quick reference sheet · Biology · keep handy for revision

1. Plant organs & meristem tissue

Root: anchors the plant; absorbs water & mineral ions (root hair cells give a large surface area).

Stem: supports the plant, holds leaves up to the light; transports substances between roots and leaves.

Leaf: main site of photosynthesis; has stomata for gas exchange and water loss.

Meristem tissue is found at the tips of roots and shoots. It is made of unspecialised cells that divide continuously by mitosis, producing new cells throughout the plant's life. These new cells then **differentiate** into specialised cells (e.g. root hair, xylem, phloem) — this is how a plant keeps growing.

2. Xylem & phloem: the transport tissues

Tissue	Transports	Direction	Cells
Xylem	Water & dissolved mineral ions	One way: up (roots → leaves)	Dead, no cytoplasm, walls strengthened with lignin
Phloem	Dissolved sugars (mainly sucrose)	Both ways (up & down)	Living cells

Moving sugars around the plant through phloem is called **translocation**.

3. Transpiration: what it is & the pathway

Transpiration is the loss of water vapour from a plant's leaves, mainly through the stomata, by evaporation and diffusion. As water is lost, more is drawn up from the roots to replace it — this continuous flow is the **transpiration stream**.

The pathway, step by step:

1. Water is absorbed from the soil by **root hair cells**, by osmosis
2. It moves across the root and into the **xylem**
3. It travels up the xylem, through the stem, into the leaves
4. It evaporates from cells inside the leaf into the air spaces, then diffuses out through tiny pores called **stomata**
5. Each stoma is surrounded by a pair of **guard cells**, which control whether it is open or closed

4. Factors affecting the rate of transpiration

Factor	Increases the rate when...	Why
Temperature	higher	molecules have more energy, so evaporate & diffuse faster
Humidity	lower	bigger difference in water vapour concentration inside/outside the leaf

Air movement (wind)	higher	removes water vapour from around the leaf, keeping the gradient steep
---------------------	--------	---

Light intensity	higher	more stomata open (for gas exchange), so more water can escape
-----------------	--------	--

A **potometer** measures the rate of water uptake by a cut shoot (used as an estimate of transpiration rate) by timing how far an air bubble moves along a capillary tube.